Sentiment Analysis - Technical Memo

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1. Overview

Cantonese is an under-resourced language in natural language processing, with less representation than other languages due to limited datasets and evaluation benchmarks. The CantoNLU study tested BERT-based models for Cantonese sentiment classification but did not test traditional machine-learning models. Our project investigates whether simpler models can achieve results comparable to BERT models, using 11,997 Cantonese restaurant reviews labelled as positive, negative, or neutral. We're comparing a majority-class baseline, character logistic regression, and fine-tuned BERT models. The goal is to measure discrepancies in performance and computational complexity between the approaches.

2. Working Data Pipeline

This project uses a sentiment dataset containing 11,997 reviews, divided into 9,999 training, 999 validation, and 999 test samples. The dataset refers to the three sentiment categories as: smile (positive), cry (negative), and ok (neutral). The data is converted to JSONL format with 'convert_to_jsonl.py', then loaded and prepared for validation with 'load_data.py'. The resulting train.jsonl, valid.jsonl, and test.jsonl files are used for model training and evaluation.

3. Baseline Implementations

The baseline models implemented and tested for Cantonese sentiment classification were a majority-class baseline and logistic regression. The majority-class baseline predicts the most common sentiment label, whereas the logistic regression uses character bigram features to identify patterns in Cantonese text and classify reviews by associated labels. The parameter C was tested with different regularization values. The configurations with the highest validation macro F1 score were chosen. Both models were evaluated on the validation set using macro F1 to measure their performance across all three sentiment classes.

4. Baseline Comparison

The baseline results are documented in 'baseline_table.md' in the repository. The logistic regression model achieved an F1 score of 0.7705, significantly outperforming the majority-class baseline of 0.1667. The logistic regression model also slightly outperformed the Cantonese BERT benchmark, with a score of 0.7705 compared to 0.719. It's important to note that the results are not representative of performance, as the baselines were evaluated on the validation set while BERT was evaluated on the test set. As stated before, logistic regression was tested at multiple C values, and the best validation score was recorded. For a fair comparison, the test set will be evaluated with the selected regression model.

5. Error Analysis

   The details of errors can be found in ‘error_analysis.md’ in the repository. The logistic regression baseline incorrectly classified 231 of 999 validation reviews, most of which were classified as neutral. Furthermore, 219 of the 231 incorrect predictions were mostly neutral class, with 12 misclassifications between positive and negative classes. Of the three classes, the neutral class had the lowest F1 score, 0.68, indicating that the model struggles to distinguish neutral sentiment from the other two classes. Some plausible explanations for the errors could be mixed sentiments, vague wording, or limiting character bigrams in context.

6. Start / Stop / Continue Plan

   The team should begin comparing the performance and computational costs of the BERT models against the majority-class and logistic regression baselines, specifically the Chinese, Cantonese, and multilingual models. The team should investigate whether BERT can reduce neutral class misclassifications, for which the logistic regression model accounted for 219 of the 231 errors.

   The team should stop tuning C in the logistic regression baseline beyond 3, as there is little to no improvement at larger values. Further tuning with higher values of C won’t resolve the errors, with the neutral class “ok” involved in 219 of the 231 incorrect predictions. Instead, we will focus on these errors and start fine-tuning as we use BERT models in future milestones.

   The team should continue using fixed dataset splits and macro F1 evaluation. The team should also continue to monitor and document model configurations, results, and errors to ensure a reproducible environment and fair comparisons between models.

7. Reproductibility

   To ensure reproducibility, this project uses the CanotNLU sentiment dataset. All baseline models are evaluated using the macro F1 score performance to ensure a consistent comparison between models. The GitHub repository contains data-loading scripts, data-processing scripts, model implementations, and documentation of the required dependencies and execution steps. The files ‘baseline_table.md’ and ‘error_analysis.md’ provide model comparisons and misclassification analysis. The models can be reproduced by running ‘naive_baseline.py’ and ‘lr_baseline.py’ from the ‘data/sentiment/’ directory.